

Nate Woodward

LC T P Spring symposium: Theoretical Physics and AI

Post-RL Performance: Synthetic & TFP-Bench

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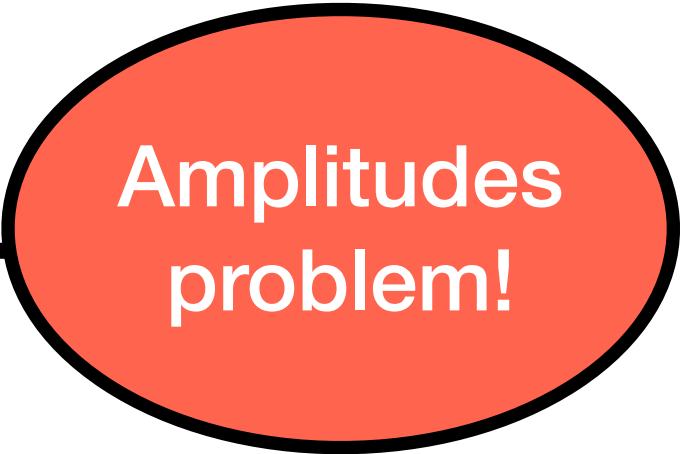
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(a) **Synthetic Dataset Performance**

Stage	Easy		Medium		Hard	
	Succ	Bo5	Succ	Bo5	Succ	Bo5
Base	40.2	67.5	26.2	50.0	0.0	0.0
1. + RL on E	54.2	71.2	44.0	62.5	2.0	5.0
Δ	+13.7	+3.7	+17.8	+12.5	+2.0	+5.0

(d) TP-Bench Performance.

Stage	Success Rate by Level					Overall	
	L1	L2	L3	L4	L5	Succ	Bo5
Base	67.5	58.5	3.6	0.0	0.0	23.5	36.8
+ RL on E	70.0	83.1	3.6	4.3	0.0	30.5	38.6
Δ	+2.5	+24.6	0.0	+4.3	0.0	+7.0	+1.8



Amplitudes
problem!



Post-RL Performance: Synthetic & TP-Bench

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Amplitudes problem!

Post-RL Performance: ArXiv & Pedagogy

(b) ArXiv Performance.

Stage	hep-th	hep-ph	gr-qc	math-ph	quant-ph	class-ph	Other	Overall
Base	12.4	9.1	23.8	10.0	22.6	16.7	15.4	16.6
+ RL on E	18.9	18.2	32.3	18.5	28.9	28.3	24.2	24.6
Δ	+6.5	+9.1	+8.5	+8.5	+6.3	+11.6	+8.8	+8.0

(c) QFT Pedagogy Performance.

Stage	Textbooks	Exercise Books	MIT OCW	Overall
Base	7.4	8.1	6.4	7.8
+ RL on E	10.4	16.8	9.3	14.2
Δ	+3.0	+8.7	+2.9	+6.4